Linear Algebra — Lecture 5

Change of Coordinate Matrix, Dual Spaces, the Rank of a Matrix

Friedberg–Insel–Spence, *Linear Algebra* 4e, §2.5, §2.6, §3.1–§3.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §2.2–§2.3, §4.4, §4.6–§4.7

Topics. Coordinates and the coordinate mapping, the change of coordinate matrix, similar matrices, dual spaces and the transpose, and the rank of a matrix.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Coordinates, mapping	§4.4 (p. 216–221)	§2.5 (p. 110–111)
Change of coordinate matrix	§4.7 (p. 239–242)	§2.5 (p. 111–112)
Similar matrices	—	§2.5 (p. 112–115)
Linear functionals	—	§2.6 (p. 119–121)
Dual basis, transpose	—	§2.6 (p. 120–122)
The double dual	—	§2.6 (p. 121–123)
Elementary matrices	§2.2 (p. 102–108)	§3.1 (p. 148–150)
The rank of a matrix	§4.6 (p. 230–236)	§3.2 (p. 152–159)

Two questions that drive the whole lecture: (1) *After a change of basis, how do the coordinates of a vector and the matrix of an operator change?* (2) *How does the number "the dimension of the image of a matrix" behave under a change of basis, under elementary operations and under multiplication?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Coordinates and the Change of Coordinate Matrix

Lecture 3 turned an ordered basis $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ of V into a coordinate machine: every $\boldsymbol{x} \in V$ is written uniquely as $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$, and $[\boldsymbol{x}]_\beta$ is that coefficient vector. This first section does one thing: it **changes the machine** and asks how the coordinates and the matrices move with it.

Definition 1.1 (Coordinate vector; Lay §4.4, p. 216)**.** Let $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ be an ordered basis of V. For $\boldsymbol{x} \in V$, the scalars $c_1, \dots, c_n$ in $\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ are the **coordinates** of $\boldsymbol{x}$ relative to β, the column

$$[\boldsymbol{x}]_\beta = (c_1, \dots, c_n)^\top$$

is the **coordinate vector** of $\boldsymbol{x}$, and $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is the **coordinate mapping**.

Theorem 1.1 (Unique Representation Theorem; Lay Theorem 7, §4.4, p. 216)**.** *Let β be a basis of V. Then every $\boldsymbol{x} \in V$ has exactly one expansion*

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$$

in the vectors of β.

Proof. Existence is "β spans V"; uniqueness is "β is linearly independent": if $\sum_i c_i \boldsymbol{b}_i = \sum_i d_i \boldsymbol{b}_i$, then $\sum_i (c_i - d_i)\boldsymbol{b}_i = \boldsymbol{0}$, so all $c_i - d_i = 0$. □

Theorem 1.2 (The coordinate mapping is an isomorphism; Lay Theorem 8, §4.4, p. 219)**.** *Let β be an ordered basis of an n-dimensional space V. Then $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is an **isomorphism** from V onto $\mathbb{F}^n$.*

Proof. Linearity: coordinates are determined by the basis, so if $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$ and $\boldsymbol{y} = \sum_i d_i \boldsymbol{b}_i$, then $\boldsymbol{x} + \boldsymbol{y} = \sum_i (c_i + d_i)\boldsymbol{b}_i$, that is, coordinates add componentwise; the same holds for scalar multiples. Injectivity: if $[\boldsymbol{x}]_\beta = \boldsymbol{0}$, then $\boldsymbol{x} = \sum_i 0 \cdot \boldsymbol{b}_i = \boldsymbol{0}$. Surjectivity: given $(c_1, \ldots, c_n)^\top$, take $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$. □

Definition 1.2 (Change of coordinate matrix; Friedberg §2.5, p. 110–111)**.** Let β, β' be two ordered bases of V. The matrix

$$Q = [I_V]^{\beta}_{\beta'}$$

is the **change of coordinate matrix** that changes β'-coordinates into β-coordinates; by the definition of a matrix representation, its **j-th column** is the coordinate vector $[\boldsymbol{b}'_j]_\beta$ of the j-th vector of β'.

Theorem 1.3 (How coordinates change; Friedberg Theorem 2.22, §2.5, p. 111)**.**

(a) Q is invertible;
(b) for every $\boldsymbol{v} \in V$, $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$.

Proof. (a) I_V is invertible, and "invertible if and only if its matrix representation is invertible" (Lecture 4 notes, Theorem 4.3) makes $Q = [I_V]^{\beta}_{\beta'}$ invertible.

(b) By "the matrix acting on coordinates gives the coordinates of the image" (Lecture 4 notes, Theorem 2.2),

$$[\boldsymbol{v}]_\beta = [I_V(\boldsymbol{v})]_\beta = [I_V]^{\beta}_{\beta'}[\boldsymbol{v}]_{\beta'} = Q[\boldsymbol{v}]_{\beta'}.$$ □

Remark 1.1 (Three sentences about Q)**.**

- **What the columns are**: the j-th column of Q is $[\boldsymbol{b}'_j]_\beta$, that is, "what the j-th new basis vector looks like in the old basis". So Q is never found by solving equations: expand each vector of β' in β.
- **Which side to multiply on**: $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$ turns β'-coordinates into β-coordinates by **left** multiplication by Q; conversely $[\boldsymbol{v}]_{\beta'} = Q^{-1}[\boldsymbol{v}]_\beta$. To remember the direction, read $Q = [I_V]^{\beta}_{\beta'}$: subscript in, superscript out. The same sense of direction governs the change of basis of an operator: $[T]_{\beta'}$ needs both ends in the new basis, hence Q on the left and Q^{-1} on the right, that is $Q^{-1}[T]_\beta Q$.
- **Two special cases**: if $\beta' = \beta$ then $Q = I_n$; if $V = \mathbb{F}^n$ and β is the standard basis, the columns of Q are simply the vectors of β' themselves, because coordinates in the standard basis are the entries.

Example 1.1 (Computing Q; Friedberg Example 1, §2.5, p. 112)**.** In $\mathbb{R}^2$ take $\beta = \{(1,1), (1,-1)\}$ and $\beta' = \{(2,4), (3,1)\}$. Find the matrix Q that changes β'-coordinates into β-coordinates, and compute $[(2,4)]_\beta$.

Solution. From

$$(2,4) = 3(1,1) - 1(1,-1), \qquad (3,1) = 2(1,1) + 1(1,-1)$$

we get

$$Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}, \qquad [(2,4)]_\beta = Q \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$ □

Approach. The columns of Q are the coordinate vectors of the vectors of β' in the basis β, so each of $(2,4)$ and $(3,1)$ has to be written as a combination of $(1,1)$ and $(1,-1)$; these are two small linear systems, and with these numbers the sum and the difference give the answer at once.

Definition 1.3 (Linear operator; Friedberg §2.5, p. 111)**.** Let V be a vector space over $\mathbb{F}$. A linear transformation T whose domain and codomain coincide, that is $T : V \to V$, is a **linear operator** on V; the set of all such T is written $\mathcal{L}(V)$. Two operators can be composed; the distributive, associative and identity laws for composition are exactly the four laws of Lecture 4 notes, Theorem 1.2, in the language of structure, $\mathcal{L}(V)$ is a (non-commutative) algebra.

Theorem 1.4 (Changing the basis of an operator; Friedberg Theorem 2.23, §2.5, p. 112)**.** *Let T be a linear operator on a finite-dimensional space V, let β, β' be ordered bases of V, and let Q change β'-coordinates into β-coordinates. Then*

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q.$$

Proof. Write $I = I_V$. Since $T = IT = TI$, the matrix of a composition (Lecture 4 notes, Theorem 2.3) gives

$$Q[T]_{\beta'} = [I]^\beta_{\beta'}[T]^{\beta'}_{\beta'} = [IT]^\beta_{\beta'} = [TI]^\beta_{\beta'} = [T]^\beta_\beta [I]^\beta_{\beta'} = [T]_\beta Q,$$

and multiplying on the left by Q^{-1} finishes the proof. □

Proposition 1.5 (The matrix version; Friedberg §2.5, p. 115, Corollary)**.** *Let $A \in M_{n\times n}(\mathbb{F})$, let γ be an ordered basis of $\mathbb{F}^n$, and let Q be the matrix whose j-th column is the j-th vector of γ. Then*

$$[L_A]_\gamma = Q^{-1}AQ.$$

Proof. Take $T = L_A$ and β the standard basis in Theorem 1.4: then $[T]_\beta = A$, and $Q = [I]^\beta_{\beta'}$ has the vectors of γ as its columns (the third special case in Remark 1.1), so $[L_A]_\gamma = Q^{-1}AQ$. □

Definition 1.4 (Similar matrices; Friedberg §2.5, p. 115)**.** Let $A, B \in M_{n\times n}(\mathbb{F})$. If $B = Q^{-1}AQ$ for some invertible Q, then B is **similar** to A. By Theorem 1.4, the matrices of one operator in two bases are always similar.

Remark 1.2 (Similarity is an equivalence relation)**.** $A = IAI^{-1}$ gives reflexivity; $B = Q^{-1}AQ$ implies $A = QBQ^{-1}$, symmetry; and if $B = Q^{-1}AQ$ and $C = P^{-1}BP$, then $C = (QP)^{-1}A(QP)$, transitivity. So similarity classifies operators: the matrices in one class are the same operator in different clothes.

Example 1.2 (Changing the basis in practice; Friedberg Example 2, §2.5, p. 113)**.** Let T be the operator on $\mathbb{R}^2$ defined by $T(a,b) = (3a-b,\ a+3b)$, with β, β' as in Example 1.1. Compute $[T]_{\beta'}$ and check it against the images of the basis vectors.

Solution. In the standard basis $[T]_\beta = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$ (the first column is $T(1,0) = (3,1)$, the second is $T(0,1) = (-1,3)$), and Example 1.1 gives $Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ with $Q^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$. Hence

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -2 & 2 \end{bmatrix}.$$ □

Approach. Take both roads. One is $Q^{-1}[T]_\beta Q$ from Theorem 1.4; the other goes back to the definition and expands $T(\boldsymbol{b}'_j)$ in β', to see whether the j-th column comes out the same. The first road is fast; the second is the only way to be sure the matrix is right.

2 Dual Spaces

Push the idea of a matrix representation to its limit and ask: **can all linear transformations on V be packed into one vector space?** Yes, into V^*. Duality is "measuring vectors with linear functionals", and the rank computations of the next section use exactly this measuring tool.

Definition 2.1 (Linear functional; Friedberg §2.6, p. 119)**.** Let V be a vector space over $\mathbb{F}$. A linear map $f : V \to \mathbb{F}$ (whose codomain is the one-dimensional space $\mathbb{F}$) is a **linear functional** on V.

Example 2.1 (Three linear functionals to remember; Friedberg Examples 1–3, §2.6, p. 119)**.**

- **Integral type**: on $V = C[0, 2\pi]$ fix g and set $h(\boldsymbol{x}) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$. With $g = \sin nt$ or $\cos nt$, $h(\boldsymbol{x})$ is the n-th Fourier coefficient.
- **Trace type**: on $V = M_{n\times n}(\mathbb{F})$, $f(A) = \operatorname{tr}(A)$.
- **Coordinate type**: fix an ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ of V and let $f_i(\boldsymbol{x}) = a_i$, where $[\boldsymbol{x}]_\beta = (a_1, \dots, a_n)^\top$. The map f_i is the i-th **coordinate function**; the point is that $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Definition 2.2 (Dual space; Friedberg §2.6, p. 119)**.** The space $\mathcal{L}(V, \mathbb{F})$ of all linear functionals on V is the **dual space** of V, written V^*. If $\dim V = n$, then $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$ (Lecture 4 notes, Proposition 4.8) gives

$$\dim V^* = \dim V.$$

Theorem 2.1 (The dual basis; Friedberg Theorem 2.24, §2.6, p. 120)**.** *Let V be n-dimensional with ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, let f_i be the i-th coordinate function and $\beta^* = \{f_1, \dots, f_n\}$. Then β^* is an ordered basis of V^*, and for every $f \in V^*$,*

$$f = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\, f_i.$$

Proof. First show that $g = \sum_i f(\boldsymbol{x}_i) f_i$ equals f: for each basis vector $\boldsymbol{x}_j$,

$$g(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i) f_i(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\delta_{ij} = f(\boldsymbol{x}_j),$$

and two linear functionals that agree on a basis are equal (Lecture 3 notes, Theorem 3.1). So β^* spans V^*; since $|\beta^*| = n = \dim V^*$, it is a basis. □

Definition 2.3 (Dual basis; Friedberg §2.6, p. 120)**.** The ordered basis $\beta^* = \{f_1, \dots, f_n\}$ of Theorem 2.1 is the **dual basis** of β; its defining property is $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Example 2.2 (Computing a dual basis; Friedberg Example 4, §2.6, p. 120)**.** Let $\beta = \{(2, 1), (3, 1)\}$ be a basis of $\mathbb{R}^2$ and find the dual basis f_1, f_2.

Solution. For f_1, write $f_1(x, y) = ax + by$; then

$$2a + b = 1, \qquad 3a + b = 0,$$

so $a = -1$, $b = 3$, i.e. $f_1(x, y) = -x + 3y$; similarly $f_2(x, y) = x - 2y$. □

Approach. The conditions $f_i(\boldsymbol{b}_j) = \delta_{ij}$ are four equations whose unknowns are the values of f_i on the standard basis. Writing $\boldsymbol{b}_j$ in terms of $\boldsymbol{e}_1, \boldsymbol{e}_2$ turns them into familiar linear systems; alternatively, read the answer off directly: f_1 must take the value 1 at $(2, 1)$ and 0 at $(3, 1)$.

Definition 2.4 (Transpose; Friedberg, §2.6, p. 121)**.** Let V, W be vector spaces over $\mathbb{F}$ and let $T : V \to W$ be linear. Define $T^\top : W^* \to V^*$ by

$$T^\top(g) = g \circ T,$$

that is, $T^\top(g)(\boldsymbol{x}) = g(T(\boldsymbol{x}))$ for all $g \in W^*$ and $\boldsymbol{x} \in V$. The map $T^\top$ is called the **transpose** of T.

Theorem 2.2 (Transpose; Friedberg Theorem 2.25, §2.6, p. 121)**.** *Let V, W be finite-dimensional with bases β, γ, and let $T : V \to W$ be linear. Then $T^\top$ is linear and*

$$[T^\top]_{\gamma^*}^{\beta^*} = \big([T]_\beta^\gamma\big)^\top.$$

Proof. The linearity of $T^\top$ follows from the compatibility of composition with addition and scalar multiplication. Let $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, $\gamma = \{\boldsymbol{y}_1, \dots, \boldsymbol{y}_m\}$ and $A = [T]_\beta^\gamma$, so that $T(\boldsymbol{x}_i) = \sum_k A_{ki}\boldsymbol{y}_k$. Expanding $T^\top(g_j)$ in β^* by Theorem 2.1, its i-th coefficient is

$$(T^\top(g_j))(\boldsymbol{x}_i) = g_j(T(\boldsymbol{x}_i)) = \sum_k A_{ki} g_j(\boldsymbol{y}_k) = A_{ji},$$

because $g_j(\boldsymbol{y}_k) = \delta_{jk}$; this is the entry in row i, column j of $A^\top$. □

Remark 2.1 (The transpose of a transformation has the transposed matrix)**.** Theorem 2.2 says: **the matrix of the transpose of a transformation is the transpose of its matrix**, with each side taken in the corresponding dual bases. Its use is to translate "left or right multiplication" into "transpose": the map $g \mapsto g \circ T$ and the map $A \mapsto A^\top$ are the same thing.

Example 2.3 (A transpose computed; Friedberg Example 5, §2.6, p. 122)**.** Let $T : P_1(\mathbb{R}) \to \mathbb{R}^2$ be $T(p(x)) = (p(0), p(2))$ with β, γ the standard bases. Then $[T]_\beta^\gamma = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$ and

$$[T^\top]_{\gamma^*}^{\beta^*} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \big([T]_\beta^\gamma\big)^\top.$$

Definition 2.5 (The element $\hat{\boldsymbol{x}}$ of the double dual; Friedberg §2.6, p. 121)**.** Let V be a vector space and $\boldsymbol{x} \in V$. Define $\hat{\boldsymbol{x}} : V^* \to \mathbb{F}$ by

$$\hat{\boldsymbol{x}}(f) = f(\boldsymbol{x}) \qquad (\text{for every } f \in V^*),$$

that is: hold $\boldsymbol{x}$ fixed and let the input be a functional, $\hat{\boldsymbol{x}}$ reads f and returns the number $f(\boldsymbol{x})$. It is linear in f, so $\hat{\boldsymbol{x}} \in V^{**}$.

Lemma 2.3 (The lemma behind the double dual; Friedberg §2.6, p. 122)**.** *Let V be finite-dimensional and $\boldsymbol{x} \in V$. If $\hat{\boldsymbol{x}}(f) = 0$ for every $f \in V^*$, then $\boldsymbol{x} = \boldsymbol{0}$.*

Proof. If $\boldsymbol{x} \neq \boldsymbol{0}$, extend it to an ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ with $\boldsymbol{x}_1 = \boldsymbol{x}$ and take f_1 from the dual basis; then $\hat{\boldsymbol{x}}(f_1) = f_1(\boldsymbol{x}) = 1 \neq 0$. □

Theorem 2.4 (The double dual; Friedberg Theorem 2.26, §2.6, p. 122)**.** *Let V be finite-dimensional and define $\psi : V \to V^{**}$ by $\psi(\boldsymbol{x}) = \hat{\boldsymbol{x}}$. Then ψ is an isomorphism.*

Proof. Linearity: $f(\boldsymbol{x} + \boldsymbol{y}) = f(\boldsymbol{x}) + f(\boldsymbol{y})$ for every f, hence $\widehat{\boldsymbol{x} + \boldsymbol{y}} = \hat{\boldsymbol{x}} + \hat{\boldsymbol{y}}$, and similarly for scalar multiples. Injectivity: Lemma 2.3. So ψ is injective, and $\dim V^{**} = \dim V^* = \dim V$ makes it surjective as well. □

Proposition 2.5 (The converse question for dual bases; Friedberg §2.6, p. 123, Corollary)**.** *Let V be finite-dimensional. Then every ordered basis of V^* is the dual basis of some basis of V.*

Proof. Let $\{f_1, \dots, f_n\}$ be an ordered basis of V^* and let $\{F_1, \dots, F_n\}$ be its dual basis in $V^{**} = (V^*)^*$, so that $F_i(f_j) = \delta_{ij}$. The map $\psi : V \to V^{**}$ of Theorem 2.4 is onto, so there are $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n \in V$ with $\hat{\boldsymbol{x}}_i = F_i$. Then for all i, j,

$$f_j(\boldsymbol{x}_i) = \hat{\boldsymbol{x}}_i(f_j) = F_i(f_j) = \delta_{ij},$$

and ψ^{-1} carries a basis to a basis, so $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ is an ordered basis of V; hence $\{f_1, \dots, f_n\}$ is its dual basis. □

Remark 2.2 (Natural and not natural)**.** The assignment $\beta \mapsto \beta^*$ depends on β: change the basis and the whole dual basis changes (hence "V is isomorphic to V^*" is true, but no isomorphism is independent of a basis). The map ψ of Theorem 2.4, by contrast, **uses no basis at all**: it is the "natural" isomorphism, which is why we identify V with V^{**} but never with V^*.

3 The Rank of a Matrix

The last part of the lecture is **one number**: $\operatorname{rank}(A)$. It is "how many dimensions the image of A has", it is "how many columns of A are really independent", and it is "how many pivots a row reduction leaves". The task of this section is to prove that these three sentences say the same thing, and to see how the number behaves under a change of basis and under multiplication.

Definition 3.1 (Elementary operations and elementary matrices; Friedberg §3.1, p. 148–150)**.** An **elementary row operation** on a matrix is one of: interchanging two rows; multiplying a row by a nonzero scalar; adding a multiple of one row to another. Column operations are defined the same way. An $m \times m$ matrix obtained from I_m by one elementary row operation is an **elementary matrix**.

Theorem 3.1 (Elementary matrices realise elementary operations; Friedberg Theorem 3.1, §3.1, p. 149)**.** *Let $A \in M_{m\times n}(\mathbb{F})$ and suppose B is obtained from A by one elementary row (column) operation. Then there is an $m \times m$ ($n \times n$) elementary matrix E with*

$$B = EA \qquad (\textit{respectively } B = AE),$$

where E is obtained from I_m (I_n) by the same operation.

Theorem 3.2 (Elementary matrices are invertible; Friedberg Theorem 3.2, §3.1, p. 150)**.** *Every elementary matrix is invertible, and its inverse is an elementary matrix of the same type.*

Proof. Let E come from I_n by one elementary row operation. Undo that step (interchange again; multiply by $1/c$; subtract the same multiple), and E goes back to I_n by an operation of the same type. By Theorem 3.1 there is an elementary matrix E' of that type with $E'E = I_n$, so E is invertible and $E^{-1} = E'$. □

Definition 3.2 (Rank; Friedberg §3.2, p. 152; Lay §4.6, p. 233)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and let $L_A : \mathbb{F}^n \to \mathbb{F}^m$ be its left-multiplication map. Define

$$\operatorname{rank}(A) = \dim \mathcal{R}(L_A) = \dim \operatorname{Col}(A),$$

the dimension of the column space of A (the span of its columns). Then the dimension theorem gives $\operatorname{rank}(A) + \operatorname{nullity}(L_A) = n$, that is $\dim \operatorname{Col}(A) + \dim \operatorname{Nul}(A) = n$.

Remark 3.1 (The range is the column space)**.** Let $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$ be the columns of A. Matrix-vector multiplication is, by definition, a combination of the columns: $A\boldsymbol{x} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$. Hence the range

$$\mathcal{R}(L_A) = \{A\boldsymbol{x} : \boldsymbol{x} \in \mathbb{F}^n\} = \{x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n : x_j \in \mathbb{F}\}$$

is the same subspace as the column space $\mathrm{Col}(A) = \mathrm{Span}\{\boldsymbol{a}_1, \dots, \boldsymbol{a}_n\}$. The two sets coincide, so their dimensions do; the second equality does not compute a second dimension.

Theorem 3.3 (Rank does not depend on the representation; Friedberg Theorem 3.3, §3.2, p. 152)**.** *Let $T : V \to W$ be linear and let β, γ be ordered bases of V, W. Then*

$$\mathrm{rank}(T) = \mathrm{rank}\big([T]_\beta^\gamma\big).$$

Proof. By Lecture 4 notes, Theorem 2.2, the maps $\boldsymbol{v} \mapsto [T(\boldsymbol{v})]_\gamma$ and $[\boldsymbol{v}]_\beta \mapsto [T]_\beta^\gamma [\boldsymbol{v}]_\beta$ are the same assignment, and the coordinate mappings are isomorphisms (Theorem 1.2), so the two images have the same dimension. □

Theorem 3.4 (Multiplying by invertible matrices preserves the rank; Friedberg Theorem 3.4, §3.2, p. 153)**.** *Let A be $m \times n$ and let P, Q be invertible of sizes $m \times m$ and $n \times n$. Then*

$$\mathrm{rank}(AQ) = \mathrm{rank}(PA) = \mathrm{rank}(PAQ) = \mathrm{rank}(A).$$

In particular (with P, Q products of elementary matrices): ***elementary row and column operations preserve the rank.***

Proof. L_Q is onto (as Q is invertible), so $\mathcal{R}(L_{AQ}) = L_A(L_Q(\mathbb{F}^n)) = L_A(\mathbb{F}^n) = \mathcal{R}(L_A)$, and taking dimensions gives $\mathrm{rank}(AQ) = \mathrm{rank}(A)$. Also $\mathrm{rank}(PA) = \mathrm{rank}((PA)^\top) = \mathrm{rank}(A^\top P^\top)$ with $P^\top$ invertible, so the result just proved applies (using $\mathrm{rank}(A) = \mathrm{rank}(A^\top)$, see Proposition 3.6). The two identities together give $\mathrm{rank}(PAQ) = \mathrm{rank}(A)$. □

Theorem 3.5 (Rank is the maximal number of independent columns; Friedberg Theorem 3.5, §3.2, p. 153)**.** *The rank of a matrix equals the maximal number of linearly independent columns, that is, the dimension of the subspace spanned by its columns.*

Proof. $\mathrm{rank}(A) = \dim \mathcal{R}(L_A)$, and $\mathcal{R}(L_A) = \mathrm{Span}\{A\boldsymbol{e}_1, \dots, A\boldsymbol{e}_n\} = \mathrm{Span}\{\boldsymbol{a}_1, \dots, \boldsymbol{a}_n\}$ (Lecture 4 notes, Theorem 2.1(b)), which is exactly the column space. □

Proposition 3.6 (Row rank equals column rank; Friedberg §3.2, p. 158, Corollary 2)**.** *For every $m \times n$ matrix A,*

$$\mathrm{rank}(A^\top) = \mathrm{rank}(A),$$

and $\mathrm{rank}(A)$ *is also the maximal number of linearly independent rows; the row space and the column space have the same dimension.*

Theorem 3.7 (The normal form for a given rank; Friedberg Theorem 3.6, §3.2, p. 155)**.** *Let A be $m \times n$ of rank r. Then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into*

$$D = \begin{bmatrix} I_r & O_1 \\ O_2 & O_3 \end{bmatrix},$$

where O_1, O_2, O_3 are zero matrices (that is $D_{ii} = 1$ for $i \le r$ and all other entries 0).

Proposition 3.8 (Two consequences of the normal form; Friedberg §3.2, p. 158–159)**.**

(a) A of rank r can be written as $D = BAC$ with B, C invertible;
(b) every invertible matrix is a product of elementary matrices.

Proof. (a) Every elementary operation is a left or right multiplication by an elementary matrix (Theorem 3.1); multiply those matrices in order to get B, C, which are invertible by Theorem 3.2 and by "a product of invertible matrices is invertible".

(b) If A is invertible then $\mathrm{rank}(A) = n$, so the normal form is $D = I_n$; by (a), $I_n = BAC$, hence $A = B^{-1}C^{-1}$, and B^{-1}, C^{-1} are products of elementary matrices. □

Algorithm 3.1 (The matrices P and Q in the normal form; Friedberg §3.2, p. 155–158)**.**

Input: *an $m \times n$ matrix A.*
Output: *invertible $P \in M_{m\times m}$ and $Q \in M_{n\times n}$ with $A = P\Sigma Q$, where Σ has I_r in its upper left corner and zeros elsewhere, $r = \operatorname{rank}(A)$.*

1. *For $k = 1, 2, \dots, m$ in turn:*
   (a) *if row k has already been cleared to a zero row, append k to the sequence ℓ and move on;*
   (b) *take the leftmost nonzero column of row k, append its column index to the sequence j, and divide row k by the entry in that column so that the pivot becomes 1;*
   (c) *use row k to clear that pivot column in every other row, including rows already handled.*
2. *For $i = 1, 2, \dots, r$, move the i-th nonzero row to position i; the result is the reduced echelon form R.*
3. *The first r columns of P are columns $j_1, \dots, j_r$ of A, its last $m-r$ columns are $\boldsymbol{e}_{\ell_1}, \dots, \boldsymbol{e}_{\ell_{m-r}}$.*
4. *The first r rows of Q are the first r rows of R; in increasing order of the column index, add the row $\boldsymbol{e}_j^\top$ for each nonpivot column j.*

Proof. Let E be the product of the row operations of steps 1 and 2, so that $R = EA$. Every pivot column of R carries a single 1 and zeros elsewhere, hence

$$E\boldsymbol{a}_j = \sum_{i=1}^{r} R_{ij}\boldsymbol{e}_i, \qquad \text{that is} \qquad \boldsymbol{a}_j = \sum_{i=1}^{r} R_{ij}\boldsymbol{c}_i,$$

where $\boldsymbol{c}_i = E^{-1}\boldsymbol{e}_i$ is column j_i of A. So every column of A is a linear combination of the first r columns of P, with coefficients taken from the first r rows of Q; and $P\Sigma Q$ is the product of those two blocks, hence $A = P\Sigma Q$.

P is invertible. Write $\boldsymbol{r}_j$ for row j of A. In the elimination, row j is interchanged, scaled by a nonzero factor and reduced by multiples of pivot rows, so it remains a linear combination of $\boldsymbol{r}_j$ and of pivot rows; and $\boldsymbol{r}_j$ occurs only in the expression for row j, so its coefficient never vanishes. Let the original indices of the pivot rows be $s_1 < \cdots < s_r$. When step 1(a) records k as ℓ_t, row k is already a zero row, so for some $c_t \neq 0$,

$$c_t\,\boldsymbol{r}_{\ell_t} + \sum_q c_{tq}\,\boldsymbol{r}_{s_q} = \boldsymbol{0},$$

and dividing by c_t gives $\boldsymbol{r}_{\ell_t} + \sum_q d_{tq}\,\boldsymbol{r}_{s_q} = \boldsymbol{0}$.

Put $\boldsymbol{y}_t = \boldsymbol{e}_{\ell_t} + \sum_q d_{tq}\boldsymbol{e}_{s_q}$; then $\boldsymbol{y}_t^\top A = \boldsymbol{0}$, and the ℓ_t-th entry of $\boldsymbol{y}_t$ equals 1. If $\boldsymbol{e}_{\ell_t} \in \operatorname{Col}(A)$, say $\boldsymbol{e}_{\ell_t} = A\boldsymbol{x}$, then multiplying on the left by $\boldsymbol{y}_t^\top$ gives $1 = (\boldsymbol{y}_t^\top A)\boldsymbol{x} = 0$, a contradiction; hence $\boldsymbol{e}_{\ell_t} \notin \operatorname{Col}(A)$. Moreover, if $\sum_t \alpha_t \boldsymbol{e}_{\ell_t} = A\boldsymbol{x}$, then multiplying on the left by $\boldsymbol{y}_u^\top$ gives $\alpha_u = 0$ for every u, so no nontrivial linear combination of these $\boldsymbol{e}_{\ell_t}$ lies in $\operatorname{Col}(A)$.

Together these two families make m vectors. If some linear combination of them vanishes, the part built from the standard vectors lies in $\operatorname{Col}(A)$, so by the previous paragraph its coefficients vanish; what remains is a combination of pivot columns, which vanishes only trivially. Hence the m vectors are linearly independent and form a basis of $\mathbb{F}^m$, so P is invertible.

Q is invertible. Its first r rows form I_r on the pivot columns, while the last $n - r$ rows $\boldsymbol{e}_j^\top$ vanish there. If a combination of these rows is zero, the pivot columns force the coefficients of the first r rows to vanish, and the nonpivot columns then force the rest to vanish. □

Theorem 3.9 (The Rank Theorem; Lay Theorem 14, §4.6, p. 233)**.** *Let A be $m \times n$. Then the column space and the row space have the same dimension, this common dimension is* $\operatorname{rank}(A)$*, it equals the number of pivot positions of A, and*

$$\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n.$$

Proof. The identity $\operatorname{rank}(A)+\dim\operatorname{Nul}(A)=n$ is the dimension theorem applied to $L_A:\mathbb{F}^n\to\mathbb{F}^m$. For "$\operatorname{rank}(A)$ = the number of pivots": row reduction preserves the rank (Theorem 3.4), while in echelon form the pivot columns are linearly independent and every other column is a combination of them, so the rank is the number of pivots (Theorem 3.5). Row rank equals column rank is Proposition 3.6. □

Theorem 3.10 (Monotonicity of the rank; Friedberg Theorem 3.7, §3.2, p. 159)**.** *Let $T:V\to W$, $U:W\to Z$ be linear and let A,B be matrices for which AB is defined. Then*

$$\operatorname{rank}(UT)\le\operatorname{rank}(U),\qquad \operatorname{rank}(UT)\le\operatorname{rank}(T),$$

$$\operatorname{rank}(AB)\le\operatorname{rank}(A),\qquad \operatorname{rank}(AB)\le\operatorname{rank}(B).$$

Proof. Because

$$\mathcal{R}(UT)=U\big(\mathcal{R}(T)\big)\subseteq\mathcal{R}(U),$$

taking dimensions gives the first inequality. (b): put $A'=[U]_\gamma^\beta$ and $B'=[T]_\alpha^\beta$; by Lecture 4 notes, Theorem 2.3, $[UT]_\alpha^\gamma=A'B'$, hence by Theorem 3.3

$$\operatorname{rank}(UT)=\operatorname{rank}(A'B')\le\operatorname{rank}(B')=\operatorname{rank}(T).$$

(c)(d) are the matrix versions of (a)(b); for (d) use the transpose, $\operatorname{rank}(AB)=\operatorname{rank}(B^\top A^\top)\le\operatorname{rank}(B^\top)=\operatorname{rank}(B)$. □

Remark 3.2 (The Invertible Matrix Theorem in rank language; Lay §4.6, p. 235)**.** For an $n\times n$ matrix A, each of the following is equivalent to "A is invertible": the columns of A form a basis of $\mathbb{R}^n$; $\operatorname{Col}(A)=\mathbb{R}^n$; $\dim\operatorname{Col}(A)=n$; $\operatorname{rank}(A)=n$; $\operatorname{Nul}(A)=\{\mathbf{0}\}$; $\dim\operatorname{Nul}(A)=0$. They continue the Invertible Matrix Theorem of Lecture 4, merely replacing the language of "$A\boldsymbol{x}=\boldsymbol{b}$" by the language of rank.

4 Reading the Dual Space with Row Vectors

This section rewrites the abstract objects of section 2 as concrete matrix arithmetic: vectors become columns, functionals become rows, and the dual basis, the transpose and the double dual all turn into things one can compute. The readings below come from joining the definitions of Friedberg §2.6 with Lay's column-vector language; neither book states them as theorems, and the only results used here are those already proved in this lecture.

Remark 4.1 (A linear functional is a row vector)**.** Work in $\mathbb{F}^n$ with vectors written as columns. Let $f:\mathbb{F}^n\to\mathbb{F}$ be linear and take the standard basis $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n$. Since $\boldsymbol{x}=x_1\boldsymbol{e}_1+\cdots+x_n\boldsymbol{e}_n$,

$$f(\boldsymbol{x})=x_1f(\boldsymbol{e}_1)+\cdots+x_nf(\boldsymbol{e}_n)=r\boldsymbol{x},\qquad r=\big(f(\boldsymbol{e}_1),\dots,f(\boldsymbol{e}_n)\big),$$

with r a single row. Conversely every row defines a linear functional, so $(\mathbb{F}^n)^*$ is exactly the set of all rows, and the equality of dimensions is immediate. This uses only the pairing "$1\times n$ times $n\times 1$", no inner product; writing $f(\boldsymbol{x})=\langle\boldsymbol{a},\boldsymbol{x}\rangle$ needs an inner product, and then $\boldsymbol{a}=r^\top$.

On an abstract space V the same computation needs a basis first: with $\beta=\{\boldsymbol{b}_1,\dots,\boldsymbol{b}_n\}$, put the values of f on the basis vectors in one row and call it $[f]_{\beta^*}$; then $f(\boldsymbol{x})=[f]_{\beta^*}[\boldsymbol{x}]_\beta$, where $[\boldsymbol{x}]_\beta$ is the coordinate vector defined in the first section of this lecture. A change of basis changes this row as well (Remark 4.4).

Remark 4.2 (The dual basis is the rows of the inverse matrix)**.** Let $\beta=\{\boldsymbol{b}_1,\dots,\boldsymbol{b}_n\}$ and let P have these vectors as its columns. The condition $f_i(\boldsymbol{b}_j)=\delta_{ij}$ says that the i-th row times the j-th column is δ_{ij}, and the i-th row of P^{-1} times the j-th column of P is exactly $(P^{-1}P)_{ij}=\delta_{ij}$.

Hence $f_1, \ldots, f_n$ are the n rows of P^{-1}. More precisely, the statement is about coordinate rows: the coordinate row of f_i relative to the **standard dual basis** (the coordinate functionals of the standard basis) is the i-th row of P^{-1}, and $f_i(\boldsymbol{x})$ is the pairing of that row with $\boldsymbol{x}$.

The coordinates come out at the same time: if $\boldsymbol{x} = P\boldsymbol{c}$ then $\boldsymbol{c} = P^{-1}\boldsymbol{x}$, and its i-th entry is $f_i(\boldsymbol{x})$. Example 2.2 finds the dual basis by solving equations; this reading gives it in one step:

$$P = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}, \qquad P^{-1} = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix},$$

whose rows are $f_1(x, y) = -x + 3y$ and $f_2(x, y) = x - 2y$, matching Example 2.2.

Remark 4.3 (The transpose moves the multiplication from left to right)**.** For $T(\boldsymbol{x}) = A\boldsymbol{x}$ the transpose is $T^\top(g) = g \circ T$. Write g as a row r; then

$$(g \circ T)(\boldsymbol{x}) = r(A\boldsymbol{x}) = (rA)\boldsymbol{x},$$

so $T^\top$ acts on rows by right multiplication by A, which on columns is left multiplication by $A^\top$. That is where $[T^\top]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^\top$ in Theorem 2.2 comes from: left multiplication on columns becomes right multiplication on rows, and back in column coordinates it is the transpose. Turn the row back into a column, that is, put $r = \boldsymbol{y}^\top$; the same fact then reads

$$(A\boldsymbol{x})^\top \boldsymbol{y} = \boldsymbol{x}^\top (A^\top \boldsymbol{y}).$$

Here A is $m \times n$, $\boldsymbol{x} \in \mathbb{F}^n$ is the input and $\boldsymbol{y} \in \mathbb{F}^m$ lies on the codomain side; both sides are scalars. The left side sends $\boldsymbol{x}$ into $\mathbb{F}^m$ and pairs it with $\boldsymbol{y}$; the right side pulls $\boldsymbol{y}$ back to $\mathbb{F}^n$ and pairs it with $\boldsymbol{x}$. With $\boldsymbol{y}$ fixed the identity holds for every $\boldsymbol{x}$, so it is the row-vector identity $rA = (A^\top \boldsymbol{y})^\top$: move the bracket to the other side and the transpose appears there.

Composition reverses order, $(S \circ T)^\top = T^\top \circ S^\top$, matching $(BA)^\top = A^\top B^\top$, because a row meets the last-acting matrix first.

Remark 4.4 (A change of basis moves columns and rows in opposite directions)**.** Let $Q = [I_V]_{\beta'}^{\beta}$; Theorem 1.3 gives $[\boldsymbol{x}]_\beta = Q[\boldsymbol{x}]_{\beta'}$, so column coordinates are multiplied by Q on the left.

The coordinates of a functional are taken with respect to the **dual basis**. Write $[f]_{\beta^*}$ for the coordinate row of f relative to β^*: its i-th entry is the i-th coefficient in the expansion of f in β^*, so that $f(\boldsymbol{x}) = [f]_{\beta^*}[\boldsymbol{x}]_\beta$. Substituting the change-of-coordinate formula,

$$f(\boldsymbol{x}) = [f]_{\beta^*}[\boldsymbol{x}]_\beta = [f]_{\beta^*} Q [\boldsymbol{x}]_{\beta'},$$

which shows that the coordinate row of the same f relative to β'^* is

$$[f]_{\beta'^*} = [f]_{\beta^*} Q$$

row coordinates are multiplied by the same Q on the right, while column coordinates are multiplied by it on the left. A basis and its dual move in opposite directions, which is what duality means on the level of coordinates.

Remark 4.5 (Why the double dual is natural)**.** The identity $\psi(\boldsymbol{x})(f) = f(\boldsymbol{x})$ is the evaluation pairing, and its definition involves no choice of basis. Hence, for each $\boldsymbol{x} \in V$, the functional $\psi(\boldsymbol{x}) \in V^{**}$ is determined before any basis is chosen; ψ does not depend on a basis and is a natural isomorphism (Theorem 2.4).

Coordinates are needed only for computation: with a basis β and its dual β^*, the coordinate row r of f times the coordinate column of $\boldsymbol{x}$ gives $f(\boldsymbol{x}) = r\boldsymbol{x}$; under a change of basis r and the coordinate column transform in opposite directions, so the product is unchanged, which is the coordinate form of the basis-independence above.

For comparison, the map $\boldsymbol{x} \mapsto \boldsymbol{x}^\top$ in $\mathbb{F}^n$ is likewise an isomorphism $\mathbb{F}^n \to \mathbb{F}^{n*}$, but it does depend on the standard basis: the functional it gives is $\boldsymbol{v} \mapsto \boldsymbol{x}^\top \boldsymbol{v}$, where the multiplication is the **standard inner product**, equal to the dot product only in standard coordinates. Let the vectors of β be the columns of P and substitute $\boldsymbol{x} = P[\boldsymbol{x}]_\beta$, $\boldsymbol{v} = P[\boldsymbol{v}]_\beta$:

$$\boldsymbol{x}^\top \boldsymbol{v} = [\boldsymbol{x}]_\beta^\top P^\top P [\boldsymbol{v}]_\beta.$$

Thus the functional attached to the same $\boldsymbol{x}$ has coordinate row $[\boldsymbol{x}]_\beta^\top P^\top P$ relative to β^*. A change of basis makes $P^\top P$ different from I_n in general, so the map changes with the basis: this isomorphism is not natural (Remark 2.2).

5 A Worked Example

Example 5.1 (Change of basis, dual basis and rank; written for this lecture; the methods are those of Friedberg §2.5, §2.6, §3.2 and Lay §4.6, §4.7)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 0 \end{bmatrix},$$

let $\beta' = \{(1,0,0), (1,1,0), (1,1,1)\}$ be an ordered basis of $\mathbb{R}^3$, and let β be the standard basis. (1) Write the change of coordinate matrix $Q = [I]_{\beta'}^{\beta}$ and find Q^{-1}; (2) compute $[T]_{\beta'}$ as $Q^{-1}AQ$ and check it against the images of the $\boldsymbol{b}'_j$; (3) find $\operatorname{rank}(A)$ and $\dim \operatorname{Nul}(A)$, and explain why $\operatorname{rank}([T]_{\beta'}) = \operatorname{rank}(A)$; (4) find the dual basis f_1, f_2, f_3 of β' and verify $f_i(\boldsymbol{b}'_j) = \delta_{ij}$; (5) let B be the elementary matrix obtained by interchanging rows 1 and 2 of I_3, and verify that $\operatorname{rank}(BA) = \operatorname{rank}(A)$.

Solution. **(1) The change of coordinate matrix.** Since β is the standard basis, the columns of Q are the three vectors of β':

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(Q is upper triangular with all diagonal entries 1, so its inverse is read off immediately: $\boldsymbol{e}_1 = \boldsymbol{b}'_1$, $\boldsymbol{e}_2 = \boldsymbol{b}'_2 - \boldsymbol{b}'_1$, $\boldsymbol{e}_3 = \boldsymbol{b}'_3 - \boldsymbol{b}'_2$, and these coefficients, taken as columns, give Q^{-1}.)

(2) The matrix in the new basis. First

$$AQ = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 0 & 1 & 1 \end{bmatrix},$$

then multiply on the left by Q^{-1} (subtract the next row from each row):

$$[T]_{\beta'} = Q^{-1}AQ = \begin{bmatrix} -1 & -3 & -4 \\ 2 & 5 & 7 \\ 0 & 1 & 1 \end{bmatrix}.$$

(3) Rank and nullity. Row-reduce A: subtract twice the first row from the second,

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so there are two pivots and $\operatorname{rank}(A) = 2$; the dimension theorem then gives $\dim \operatorname{Nul}(A) = 3-2 = 1$. Indeed, solving $A\boldsymbol{x} = \mathbf{0}$ gives $x_2 = 0$ and $x_1 + x_3 = 0$, that is

$$\operatorname{Nul}(A) = \operatorname{Span}\{(-1, 0, 1)^\top\}.$$

For $[T]_{\beta'}$ the same thing happens: subtracting rows produces a zero row (the third row is a combination of the first two), so $\operatorname{rank}([T]_{\beta'}) = 2 = \operatorname{rank}(A)$, in agreement with Theorem 3.3.

(4) The dual basis. Write $\boldsymbol{x} = (x_1, x_2, x_3)$ in the basis β': $\boldsymbol{x} = c_1\boldsymbol{b}'_1 + c_2\boldsymbol{b}'_2 + c_3\boldsymbol{b}'_3$, and comparing entries gives $c_3 = x_3$, $c_2 = x_2 - x_3$, $c_1 = x_1 - x_2$. The coordinate functions are the dual basis:

$$f_1(\boldsymbol{x}) = x_1 - x_2, \qquad f_2(\boldsymbol{x}) = x_2 - x_3, \qquad f_3(\boldsymbol{x}) = x_3.$$

(5) Elementary operations and rank. Let B be the elementary matrix obtained by interchanging rows 1 and 2 of I_3. It is invertible (Theorem 3.2), so Theorem 3.4 gives $\operatorname{rank}(BA) = \operatorname{rank}(A) = 2$. Directly,

$$BA = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

whose first row is twice the second while the remaining two are not proportional, so the rank is still 2. □

Approach. Each part matches one theorem. (1) is "the columns of Q are the new basis vectors in the old basis"; here the old basis is the standard one, so the columns of Q are the three vectors themselves. (2) is Theorem 1.4. (3) is "rank = number of pivots" plus Theorem 3.3. (4) is "the dual basis is fixed by $f_i(\boldsymbol{b}'_j) = \delta_{ij}$", read off from the formula that solves for β'-coordinates. (5) is Theorem 3.4: elementary matrices are invertible, so left multiplication by one cannot change the rank. The whole example comes down to *a change of basis changes the look of the matrix, never its rank.*

Example 5.2 (Similarity: a basis that puts the operator into the nilpotent shape; written for these notes; method from Friedberg §2.5)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Verify that $A^2 = O$ and $\operatorname{rank}(A) = 1$, and find an invertible matrix P with $P^{-1}AP = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Solution. **(1) Kernel and range.** The range of $T(\boldsymbol{x}) = A\boldsymbol{x}$ is spanned by its nonzero column, the third, $\mathcal{R}(T) = \operatorname{Span}\{(1, 1, 0)^\top\}$; the kernel is the solution set of $x_3 = 0$, namely $\mathcal{N}(T) = \operatorname{Span}\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$. The range lies inside the kernel, which is what $A^2 = O$ says.

(2) Choosing the basis. Take $\boldsymbol{v}_1 = (1, 1, 0)^\top$ as a basis of the range; take $\boldsymbol{u}_1 = (0, 0, 1)^\top$, since $A\boldsymbol{u}_1 = (1, 1, 0)^\top = \boldsymbol{v}_1$; and take $\boldsymbol{w} = \boldsymbol{e}_1$, which lies in the kernel and is independent of $\boldsymbol{v}_1$. These three vectors are linearly independent and form a basis $\beta = (\boldsymbol{v}_1, \boldsymbol{u}_1, \boldsymbol{w})$.

(3) Transition matrix and result. The columns of P are this basis,

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad P^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix},$$

and hence

$$P^{-1}AP = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The three columns follow from $T(\boldsymbol{v}_1) = \mathbf{0}$, $T(\boldsymbol{u}_1) = \boldsymbol{v}_1$ and $T(\boldsymbol{w}) = \mathbf{0}$: they are the coordinates of $\mathbf{0}$, $\boldsymbol{e}_1$ and $\mathbf{0}$. □

Approach. $A^2 = O$ says the range sits inside the kernel, so the first two blocks of the basis are a basis of the range and its preimages, and the rest completes the kernel; the shape of the matrix is decided by how these blocks correspond. Relation to Theorem 1.4: this is operator change of basis with $V = W$, old basis standard and new basis β; A and $P^{-1}AP$ are the matrices of the same operator T in two bases, and the columns of P are the new basis vectors.

6 Supplement: notions Homework 5 uses but the lecture did not cover

Remark 6.1 (Where these come from)**.** None of the following is part of the lecture's main thread, yet Homework 5 uses them:

- **Inclusion** and **quotient maps**: Problem 16 of the sheet.
- **Exactness** and **exact sequences**: Definitions 1 and 2 of the sheet.
- The **evaluation functional** $f_c(p) = p(c)$: Problem 14 of the sheet.
- **Change of basis between two spaces**: Theorem 1.4 covers operators only; the general form is Theorem 6.1 (Exercise 8 of Friedberg, §2.5, printed p. 117–118).

Definition 6.1 (Inclusion and quotient maps; homework sheet, Problem 16)**.** Let W be a subspace of a vector space V. The **inclusion map** is

$$S : W \to V, \qquad S(\boldsymbol{x}) = \boldsymbol{x},$$

and the **quotient map** is

$$T : V \to V/W, \qquad T(\boldsymbol{v}) = \boldsymbol{v} + W.$$

Both are linear: the first is the restriction of the identity of V to W, the second is well defined by the addition and scalar multiplication of cosets.

Definition 6.2 (Exactness; homework sheet, Definition 1)**.** Let $A \xrightarrow{S} B \xrightarrow{T} C$ be linear maps of vector spaces. The composite is **exact** at B when

$$\mathcal{N}(T) = \mathcal{R}(S),$$

that is, when $T(\boldsymbol{b}) = \mathbf{0}$ holds exactly for those $\boldsymbol{b}$ that are images of some $\boldsymbol{a} \in A$.

Definition 6.3 (Exact sequence; homework sheet, Definition 2)**.** A sequence

$$\cdots \to V_{i+1} \xrightarrow{T_{i+1}} V_i \xrightarrow{T_i} V_{i-1} \to \cdots$$

is **exact** when $\mathcal{N}(T_i) = \mathcal{R}(T_{i+1})$ for every i. The symbol 0 in such a sequence denotes both the zero vector space and the zero map ($\boldsymbol{x} \mapsto \mathbf{0}$).

Definition 6.4 (Evaluation functional; homework sheet, Problem 14)**.** Let $V = P_n(\mathbb{F})$ and $c \in \mathbb{F}$. Define

$$f_c : V \to \mathbb{F}, \qquad f_c(p) = p(c).$$

It is linear: $(p + q)(c) = p(c) + q(c)$ and $(ap)(c) = a\,p(c)$.

Theorem 6.1 (Change of basis between two spaces; Friedberg §2.5, Exercise 8)**.** *Let $T : V \to W$ be linear, let β, β' be ordered bases of V and let γ, γ' be ordered bases of W. Write $Q = [I_V]^{\beta}_{\beta'}$ and $P = [I_W]^{\gamma}_{\gamma'}$. Then*

$$[T]^{\gamma'}_{\beta'} = P^{-1}[T]^{\gamma}_{\beta}Q.$$

Proof. For $\boldsymbol{v} \in V$ the definition of the matrix of a map gives $[T(\boldsymbol{v})]_\gamma = [T]^{\gamma}_{\beta}[\boldsymbol{v}]_\beta$. Express both ends in the primed bases: $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$ and $[T(\boldsymbol{v})]_\gamma = P[T(\boldsymbol{v})]_{\gamma'}$. Substituting gives $P[T(\boldsymbol{v})]_{\gamma'} = [T]^{\gamma}_{\beta}Q[\boldsymbol{v}]_{\beta'}$ for every $\boldsymbol{v}$, and $\boldsymbol{v} \mapsto [\boldsymbol{v}]_{\beta'}$ runs over all of $\mathbb{F}^n$; hence $P[T]^{\gamma'}_{\beta'} = [T]^{\gamma}_{\beta}Q$, and multiplying by P^{-1} on the left gives the formula.

When $V = W$, $\beta = \gamma$ and $\beta' = \gamma'$ we have $P = Q$, and the formula becomes $Q^{-1}AQ$, which is the similarity relation of Theorem 1.4. □

Example 6.1 (Equivalence: given A and B, find P and Q with $A = PBQ$; written for these notes; the method is Algorithm 3.1)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

Find invertible $P \in M_{2\times 2}(\mathbb{F})$ and $Q \in M_{3\times 3}(\mathbb{F})$ with $A = PBQ$.

Solution. **(1) Both to the same normal form.** Write $\Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. The pivot columns of A are 1 and 3, with no zero row; B is already in reduced echelon form with pivot columns 1 and 2 and no zero row. Read off by Algorithm 3.1 (columns of P are the pivot columns, rows of Q are the rows of the reduced form, with $\boldsymbol{e}_j^{\top}$ added for each non-pivot column):

$$A = P_A\Sigma Q_A, \qquad P_A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}, \quad Q_A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix};$$

$$B = P_B\Sigma Q_B, \qquad P_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q_B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

(2) Splicing. From $B = P_B\Sigma Q_B$ we get $\Sigma = P_B^{-1}BQ_B^{-1}$; substituting into $A = P_A\Sigma Q_A$,

$$A = P_AP_B^{-1}\,B\,Q_B^{-1}Q_A.$$

Here $P_B = I_2$, so $P = P_A$; Q_B is unit upper triangular, so its inverse just flips the sign of the two entries above the diagonal,

$$Q_B^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix},$$

and hence

$$Q = Q_B^{-1}Q_A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$ □

Approach. Equivalent matrices are the same linear map in two pairs of bases, so finding P and Q means choosing one basis for A and one for B; a single normal form acts as the hub. All four matrices are read off by Algorithm 3.1, $P_B = I_2$ saves the product for P, and Q_B being unit upper triangular lets its inverse be written down. Relation to Theorem 6.1: that theorem changes bases in one space, and here it is used twice; if B is the matrix of a map in the basis pair (β, γ), then A is the matrix of the same map in (β', γ'), with P changing the codomain basis and Q the domain basis.

Proposition 6.2 (Multiplicativity of the trace; Friedberg §2.3, Exercise 13, p. 97)**.** *Let* $A, B \in M_{n\times n}(\mathbb{F})$*. Then*

$$\operatorname{tr}(AB) = \operatorname{tr}(BA), \qquad \operatorname{tr}(A^{\top}) = \operatorname{tr}(A).$$

Hence if $B = Q^{-1}AQ$ *with* Q *invertible, then* $\operatorname{tr}(B) = \operatorname{tr}(A)$*.*

Proof. $\operatorname{tr}(AB) = \sum_i (AB)_{ii} = \sum_i \sum_k a_{ik}b_{ki} = \sum_k \sum_i b_{ki}a_{ik} = \sum_k (BA)_{kk} = \operatorname{tr}(BA)$. Transposing swaps the indices: $\operatorname{tr}(A^{\top}) = \sum_i (A^{\top})_{ii} = \sum_i a_{ii} = \operatorname{tr}(A)$. For similar matrices, $\operatorname{tr}(Q^{-1}AQ) = \operatorname{tr}\big((Q^{-1}A)Q\big) = \operatorname{tr}\big(Q(Q^{-1}A)\big) = \operatorname{tr}(A)$. □

7 Checklist and Exercises

By the end of this lecture you should be able to:

1. state the definitions of the coordinate vector and the coordinate mapping, and explain why the coordinate mapping is an isomorphism (uniqueness gives linearity, spanning gives surjectivity, independence gives injectivity);
2. write down the change of coordinate matrix $Q = [I_V]_{\beta'}^{\beta}$, say that its columns are the vectors of β' in β, and state $[\boldsymbol{v}]_{\beta} = Q[\boldsymbol{v}]_{\beta'}$ on the spot;
3. change the basis of a 2×2 operator with $Q^{-1}[T]_{\beta}Q$ and check the result against the images of the basis vectors;
4. define similar matrices, prove that similarity is an equivalence relation, and say that rank and invertibility are invariant under similarity;
5. give the three examples of linear functionals (integral, trace, coordinate) and explain why $\dim V^* = \dim V$;
6. compute the dual basis of a given basis by solving the δ_{ij} equations, and say precisely what "the two bases are dual to each other" means;
7. state $T^{\top}(g) = g \circ T$ and $[T^{\top}]_{\gamma^*}^{\beta^*} = ([T]_{\beta}^{\gamma})^{\top}$, and explain why "the matrix of the transpose is the transpose of the matrix";
8. state the double dual theorem and explain why ψ is natural while an isomorphism $V \cong V^*$ is not;
9. compute the rank of a matrix by counting pivots in a row reduction, or by counting a maximal independent set of columns, and pass between the two descriptions;
10. say that rank is preserved by multiplication by invertible matrices and by elementary operations, and state the Rank Theorem $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$;
11. use the rank inequalities (Theorem 3.10) to bound the rank of a composite or a product from above and below;
12. read a dual basis as the rows of P^{-1} with row vectors, write $[\boldsymbol{x}]_{\varepsilon} = P[\boldsymbol{x}]_{\beta}$ and $[f]_{\beta'^*} = [f]_{\beta^*}Q$, and explain why column coordinates are multiplied by Q on the left while row coordinates are multiplied by Q on the right.

Exercises (printed pages). Lay: §4.4 p. 222 (Exercises 1–13: coordinate vectors and the coordinate mapping), §4.7 p. 242 (Exercises 1–9: change-of-coordinates matrices), §4.6 p. 236 (Exercises 1–23: rank, row space, the Rank Theorem), §2.2 p. 108 (Exercises 1–21: elementary matrices and inverses), §2.3 p. 114 (exercises on the Invertible Matrix Theorem); Friedberg: §2.5 p. 116 (Exercises 2, 3, 6, 8: change of basis and similarity), §2.6 p. 123 (Exercises 2–7: dual bases and the transpose), §3.1 p. 151 (Exercises 1–8: elementary matrices), §3.2 p. 165 (Exercises 1–15: rank, normal form, rank inequalities). Each section also has Practice Problems; do those first.

How to use the two texts together:

- **Compute and prove are divided cleanly.** How to find a change of coordinate matrix, how to count a rank: Lay (§4.4, §4.7, §4.6). Why the change of basis is $Q^{-1}AQ$, why the

double dual is natural, why the rank does not depend on the representation: Friedberg (§2.5, §2.6, §3.2).

- The core chain of the lecture: *the coordinate mapping is an isomorphism* $\to$ *the change of coordinate matrix* $Q = [I_V]^{\beta}_{\beta'}$ $\to$ *the operator changes basis as* $[T]_{\beta'} = Q^{-1}[T]_{\beta}Q$ $\to$ *similarity is "the operator in different clothes"* $\to$ *the rank survives the change of clothes.* The second thread is *duality*: *linear functionals* $\to$ *the dual basis* $\to$ *the transpose* $\to$ *the double dual is natural.*
- The central statement is Theorem 3.3: it welds "the rank of an operator" to "the rank of its matrix", so that the rank is at once a geometric quantity (the dimension of the image) and a computable one (the number of pivots), and Theorem 3.4 says it is also an **invariant**.